Exercise 3

Theodosius Bekele
Data Science.

① Classify each product by product
* Expensive if price > 100
* Mid range if price between 100 and 1000
* Budget if price < 100

```
Select product_name, price,
~~From brightlearn.databases.exercise_3_products~~
Case
    when price > 1000 then 'Expensive'
    when price between 100 and 1000 then 'Mid-range'
    else 'Budget'
    end as price_category;
From brightlearn.databases.exercise_3_products;
```

Product Name	price	price_category
Laptop	1200.00	Expensive
Phone	800.00	mid-range
keyboard	45.00	Budget
Monitor	300.00	Mid-range
Mouse	25.00	Budget

② Lebel each order: * High value For Orders ≥1000
* Medium value For 500-999.99
* Low value For <500

```
Select customer_name,
    amount,
Case
    when amount >= 1000 then 'high value'
    when amount between 500 and 999.99 then 'Medium value'
    ~~when~~ else 'low value
    end as ordercategory
From brightlearn.databases.exercise_3_Products;
```

Product name	Price	Price category
Laptop	1200.00	Expensive
Phone	800.00	Mi
keyboard	45.00	
Monitor	300.00	
Mouse	25.00	

customer name	amount	Order-value category
Alice	150.00	low value
Bob	560.00	Medium value
Charlie	999.99	Medium value
Diana	45.50	low value
Ethan	1200.00	high value

② Exercise 3 — Thulasiram Kulkarni
Data Science

③ Categorize employee position: If in IT and salary > 80000 'Senior IT'
If in HR and salary > 55000 'Experience HR'
otherwise 'Staff'

```
Select emp_name,
       department,
       Salary,

Case
    when department = 'IT' and salary > 80000 then 'Senior IT'
    when department = 'HR' and salary > 55000 then 'Experienced HR'
    Else 'Staff'
    End as position_level
From brightlearn.databases.exercise_3_employees;
```

Emp_name	department	Salary	position_level
John	IT	85000	Senior IT
Sara	HR	60000	Experience HR
Mark	IT	75000	Staff
Lucy	Finance	90000	Staff
Tom	HR	55000	Staff

④ Assign a letter grade: ≥ 90: A, 80-89: B, 70-79: C, 60-69: D, <60: F

```
Select student_Name,
       Score,

Case
    when Score ≥ 90 then 'A'
    when Score between 80 and 89 then 'B'
    when Score between 70 and 79 then 'C'
    when Score between 60 and 69 then D
    else 'F'

    end as grade,

From brightlearn.databases.exercise_5_students;
```

student_name	Score	grade
Anna	92	A
Ben	76	C
Carl	59	F
David	83	B
Ella	68	D

③ Exercise 3

Thulasizwe Butheleti
Data Science

ⓐ label delivery performance. ≤ 30mins: 'Fast', 31-60 mins: 'on Time'
60mins: 'Late'

```
Select delivery_id,
       delivery_time_minutes,
Case
  when delivery_time_minutes ≤ 30 then 'Fast'
  when delivery time minuts between 31 and 60 then 'on time'
  else 'Late'
End as performance
From brightlearn.database.exercise_3_deliveries;
```

delivery_id	delivery time minute	performance
1	45	on Time
2	80	Late
3	30	Fast
4	65	Late
5	100	late

ⓒ Convert priority to labels: 3-high, 2-Medium, 1-low.

```
Select issue_type,
       Priority,
Case
  when priority = 3 then 'high'
  when priority = 2 then 'Medium'
  else 'low'
End as priority_label
From brightlearn.database.exercise_3_tickets;
```

issue type	Priority	Priority level
long issue	1	low
server down	3	high
slow system	2	Medium
email error	2	Medium
Password reset	1	low

(4) Exercise 3 — Thuvasivan Puthukodi, Data Science

(7) Calculate attendance % and classify: ≥90% - Excellent, 70-89% Good, < 75% - need improvement

```
Select student_id,
    (days_present/total_days)x 100 as attendance_percentage
Case
  when attendance_percentage ≥ 90 then (Excellent)
  when attendance_percentage between 70 and 89 then (Good)
  Else 'need improvement')
  End as attendance_status

From brightlearn.database.exercise_3_attendance;
```

student_id	attendance_percent	attendance_status
1	90	Excellent
2	60	need Improvement
3	96	Excellent
4	50	need Improvement
5	100	Excellent

(8) label stock status: 0 - out of stock, 1-5 - low stock, 5 - In stock

```
select product_id,
       stock_qty,
Case
  when stock_qty=0 then 'Out of stock')
  when stock_qty between 1-5 then 'low stock')
  ~~when stock_qty=5~~ Else 'In stock'
  end as stock_status

From brightlearn.databases.exercise_3_products_inventory;
```

Product_id	Stock_qty	stock_status
1	5	low stock
2	0	out of stock
3	25	In stock
4	10	In stock
5	3	low stock

⑤ Exercise 3 Thulasiram Buthoth
Data Science

⑨ Classify by size: 25 → Large, 10-24 - Medium, <10 - Small

```
Select subject,
       enrolled_students,
case
  when enrolled_students ≥ 25 then 'Large'
  when enrolled_students between 10 and 24 then 'Medium'
  else 'Small'
  end as class_size_category
From bright learn. database. exercise_3_classes;
```

Subject	enrolled_students	class_size_category
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

⑩ Apply discount Flag: If payment method = 'cash' and amount ≥ 200 → 'Eligible for discount'
Else not eligible.

```
Select payment_id,
       amount,
       Payment_method,
case
  when payment_method = 'cash' and amount ≥ 200 then
  (Eligible For Discount)
  , else 'Not Eligible'
  end as discount_eligibility
From bright learn. database. exercises_payments;
```

Payment id	amount	Payment method	discount eligibility
1	50.00	Card	not eligible
2	200.00	cash	Eligible for discount
3	150.00	Card	not eligible
4	75.00	PayPal	not eligible
5	300.00	cash	Eligible for discount